

Fran Menéndez Moya

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| Zaragoza, Spain | EU work authorization | Remote

SUMMARY

Software Engineer with 10+ years building platforms that serve millions of users. I work at the AI layer and across the full stack: a multi-agent delivery system I built and run, semantic search in production, and MCP tooling I made the team-wide standard. I lead architecture domain-wide, rebuild the systems holding a product back, set standards the team adopts, and mentor engineers into ownership.

TECHNICAL SKILLS

AI Engineering: Agent System Design (multi-agent orchestration, dependency-derived scheduling, schema-typed review gates, bounded retries), Model & Effort Routing, Adversarial AI Code Review, Context Engineering, MCP (tooling & design-to-code), Semantic Search (OpenSearch k-NN, vector embeddings, BM25)
Languages: TypeScript, JavaScript, HTML/CSS, SQL
Application Security: SQL injection prevention, access control & IDOR, PII exposure, security regression testing in CI
Frontend: Next.js, React, Vue.js, Angular, Astro
Backend: Node.js, NestJS, Microservices, Real-time Systems, REST, GraphQL
Data & Search: PostgreSQL, MongoDB, CouchDB, OpenSearch, Redis
Infrastructure: AWS (Cognito, Lambda, CloudFront, S3), Kubernetes, Docker, CDN, Message Queues
CI/CD & Tools: Jenkins, GitHub Actions, CircleCI, Fastlane, Git, Automated Pipelines
CMS & Commerce: PayloadCMS, Contentful (incl. custom editor apps), Headless Architecture
Mobile & Cross-Platform: React Native, Ionic, Cordova, Electron, PWAs

PROFESSIONAL EXPERIENCE

The Knot Worldwide Oct 2023 - Jul 2026

Remote

Principal Software Engineer Apr 2025 - Jul 2026

- Rebuilt the Shop e-commerce platform from scratch in one month solo (versus a three-month, multi-developer estimate). The platform handles millions of products from Amazon's Rainforest API and editorial content via Contentful, built with Next.js and React with shared component libraries, now serving hundreds of thousands of users in production.
- Built a semantic product matching system using OpenSearch with k-NN vector similarity and BM25 text relevance, banded by confidence: candidates below a 0.75 floor were discarded, the strongest matches served automatically - about 90% of throughput, with no human review - and the band in between routed to a custom review dashboard I built inside PayloadCMS. p95 query latency stayed under 50ms across 100,000+ products.
- Introduced Model Context Protocol (MCP) tooling that generates production components directly from Figma at token-exact fidelity, removing the manual design-to-code translation step; now the team-wide standard. On my own implementation work it took design-with-components from about a week to under a day, roughly 85%.
- Built and ran an agentic AI development workflow for my own production delivery: four stages, eight role-scoped agents, model routing by task, a thin-orchestrator context budget, and review roles barred from writing the code they audit. Used it to deliver a security hardening programme, where the independent step caught defects the implementing pass had missed; drove its practices into the team's process.
- Ran a full API security audit and hardening programme for the public e-commerce service: eight findings across four vulnerability classes - SQL injection, over-open collection access, PII projection and identity trust - extended by roughly nine further findings once a full-product verification pass enumerated the access surface from code. Remediation found and closed a write-side IDOR in a shared authorisation primitive covering five collections. Built the service's first automated test harness (two-tier, DB-free CI proof layer), a lint gate enforced at error level in CI and pre-push, and a regenerable access-coverage matrix that flags any loosening of access as a diff.
- Unified sign-on across five products - Bump articles, baby names, registry, shop and the native apps - so that one account replaced five separate logins.
- Built session continuity across the Bump platform and Shop boundary: an OAuth token handoff signs users into Shop with their existing Bump account on navigation, provisioning an account just-in-time when none exists.

- Root-caused cross-service authentication failures across that same boundary, isolating an edge-layer fault and a hanging Cognito session call; resolved a Kubernetes-served canonical SEO incident.
- Migrated PayloadCMS from v2 to v3 for 100,000+ products with zero downtime; query response times dropped from 850ms to 34ms through database optimization and caching architecture.
- Building on that migration, reworked caching for 2M+ weekly users: added a cache layer and set per-endpoint revalidation and staleness policies after investigating each endpoint's tolerance, on the existing CDN with no added infrastructure. The reduced origin load let the pods be right-sized down, cutting the resource allocated and billed per pod with no availability regression.
- Steadied the engineering team through a company restructuring, and argued build-capacity against target-attainment in the half-year OKR capacity review; adjudicated review disputes on evidence rather than seniority.
- Built a custom Contentful editor app for the editorial platform (React, Contentful App SDK), including its S3 + CloudFront hosting infrastructure; propagated the new content model end-to-end across three services with zero-downtime migrations.

Lead Software Engineer

Oct 2023 - Mar 2025

- Built an A/B testing system for The Bump, a platform where edge caching had made experimentation impossible: Akamai assigns a variant cookie at the edge, the app reads it to render the matching tagged build, and assignment stays cached and sticky across reloads until the cookie expires. The platform had run no experiments in over 3 years before it; 10+ have run on it since. The winning variant in an ad-layout test lifted served ad impressions 23% against control on an ad-monetised property; it shipped because engagement depth held at baseline in Mixpanel and GA4, with the new ad combination sitting outside the width-capped article column.
- Managed an 8-person cross-functional team; mentored 5 developers, resulting in 2 promotions, with 100% retention across the period.
- Took the integrated commerce platform from proposal to a live MVP, on a \$2M+ annual revenue projection.
- Cut deployment time from over 2 hours to 1 minute, taking releases from batched to daily.

MOBIKO GmbH

Aug 2020 - Sep 2023

Munich, Germany (remote)

Team Lead & Architecture

Mar 2023 - Sep 2023

- Set the technical strategy and built the microservices architecture behind the company's enterprise expansion - three global enterprise partnerships, each scoped directly with the customer and integrated per partner - processing 500k+ daily mobility events: a NestJS API gateway fronting serverless functions communicating over NATS message queues, containerized into Kubernetes with Keda-driven autoscaling and GitHub Actions deployment workflows.
- Restructured the review and release process and mentored engineers into ownership; the team grew and delivery cycles shortened over the period.

Senior Full-Stack Developer

Aug 2020 - Mar 2023

- Rebuilt the API surface on NestJS with Swagger-annotated contracts and TypeORM against PostgreSQL, cutting API response times 72% through caching and database query optimization; page loads dropped 70% in Lighthouse.
- Scaled the user base from 3,000 to 10,000+ users (233% growth) through performance optimization and strategic feature development.
- Migrated the entire product onto a new Vue platform and built the PWA intended to replace the existing multiplatform applications, on a Vuex/Flux state architecture in a class-structured TypeScript codebase.
- Automated the deployment pipeline from multi-day release cycles to continuous deployment, on Kubernetes with CircleCI running automated deploys and compatibility checks.

Hiberus Tecnología

Jul 2017 - Jul 2020

Zaragoza, Spain

Senior Full-Stack Developer

Jan 2020 - Jul 2020

- Rebuilt the checkout path for a real-time ticketing platform under La Liga match-day load, integrating the official API for 100k+ concurrent users.
- Built Afición360, a single Angular monorepo serving the front ends of multiple football clubs plus an internal admin dashboard, on a shared Flux state architecture with NestJS REST microservices split via Lerna; integrated the La Liga and AVET APIs, using websockets deliberately to throttle traffic so the external services were not overloaded.
- Built and configured Jenkins from scratch, with pipelines building and deploying every monorepo application and backend into Docker on AWS.

Full-Stack Developer

Nov 2017 - Jan 2020

- Built a platform processing 1M+ daily entries with real-time sync across 5 geographic locations, on CouchDB as an offline-capable distributed database live-syncing data spread across multiple systems, at 99.95% uptime; still in production.

- Built cross-platform Ionic applications that ran fully offline or embedded existing web apps, using Cordova for native device capabilities (bluetooth, GPS, push notifications), and compiled Angular applications into Electron to run as kiosks; automated mobile build, test and marketplace deployment with Fastlane.

Junior Developer, E-commerce

Jul 2017 - Nov 2017

- Built a NodeJS/Express service from scratch aggregating external APIs, still in production, and an AngularJS application for a multinational consuming Google ecosystem APIs; backend work on an in-house Spring/Hibernate framework over MySQL and SQL Server.

Earlier: Freelance Web Developer

2016 - Jul 2017

- Delivered a full redesign across frontend, backend, database and self-hosted deployment for a university graduation photography studio: a public interface where students booked appointments to have their graduation photographs taken, plus an administration panel managing those appointments and private events.

INDEPENDENT WORK

Instagram Checker

Live and built end to end with the same agentic workflow: it turns an Instagram data export the user supplies into a follower and following dashboard, reading only what the user provides rather than scraping.

instagram-checker-web.vercel.app

Multi-agent delivery harness

Built and run solo: it reads a tracker's own dependency edges to derive the order work runs in, scheduling what nothing blocks in parallel and everything else behind what it waits on. An independent review reads each branch against the acceptance criteria it was written from and answers in a schema the harness parses rather than in prose, and it is that parsed verdict, never the wording around it, that gates the merge; a finding goes back to the session that wrote the code as a bounded retry rather than a rewrite. Every step is appended to a typed transition stream, so what a run did, what it cost and where it stopped are read back from that record rather than from a terminal, and each role runs at the model and the effort its own failure cost argues for.

EDUCATION & LANGUAGES

Computer Engineering (Software Engineering Specialization) [Universidad de Zaragoza](#)

2012-2017

Languages

Spanish (Native) | English (Advanced, Cambridge C1)

RECOGNITIONS & AWARDS

NASA Space Apps Global Finalist & Galactic Impact Award	May 2017
100 Ideas Zaragoza, Most Innovative Technology	Sep 2017
100 Ideas Zaragoza, Best Use of the Citizen Card	Sep 2017
uCode by Adidas, Best Fan Experience Solution	Mar 2018
ImagineCode Blockchain Challenge Winner	Oct 2018
NASA Space Apps Zaragoza Local Hub Winner	Oct 2018
Google Hash Code Zaragoza Hub Winner	Feb 2019